# Are You Getting What You Pay For?

Auditing Model Substitution in LLM APIs: Why Software Tests Fail and Hardware Attestation Wins.

Based on: arxiv.org/abs/2504.04715

# The Model Substitution Problem

Economic incentives drive covert swaps. Providers may replace advertised models to cut GPU costs while maintaining pricing. Hypothesis test: Can an auditor distinguish P_spec(y|x) from P_actual(y|x)?

`TYPE 1: SCALE`

Smaller models (e.g., 8B instead of 70B)

`TYPE 2: PRECISION`

Quantized variants (e.g., FP8/INT8 vs FP16)

`TYPE 3: VERSIONING`

Fine-tuned or updated versions

`TYPE 4: FAMILY`

Entirely different model architectures

# FP8 Quantization Preserves Most Accuracy

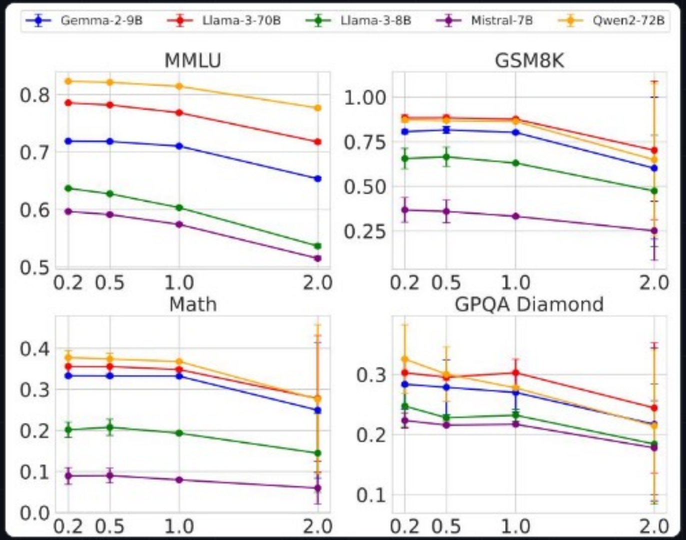
Performance drops are small (often within error bars), making substitution nearly undetectable via black-box benchmarks.

| Model Family | MMLU | GSM8K | MATH | GPQA Diamond |
|---|---|---|---|---|
| Llama-3-8B | 62.69 ± 0.18 | 61.14 ± 3.47 | 20.65 ± 3.43 | 22.62 ± 0.22 |
| Llama-3-8B-FP8 | 62.43 ± 0.26 | 60.90 ± 4.10 | 14.91 ± 2.49 | 20.14 ± 0.24 |
| Llama-3-70B | 78.05 ± 0.08 | 88.06 ± 1.44 | 35.69 ± 1.33 | 29.60 ± 0.51 |
| Llama-3-70B-FP8 | 77.88 ± 0.13 | 87.35 ± 1.24 | 35.75 ± 1.16 | 33.12 ± 0.30 |
| Qwen2-72B | 82.18 ± 0.08 | 86.72 ± 1.00 | 37.39 ± 1.41 | 29.93 ± 2.71 |
| Qwen2-72B-FP8 | 81.98 ± 0.08 | 86.82 ± 0.97 | 37.67 ± 1.39 | 31.08 ± 1.96 |
| Gemma-2-9b | 71.86 ± 0.08 | 81.80 ± 1.35 | 33.41 ± 0.28 | 28.64 ± 2.97 |
| Gemma-2-9b-FP8 | 71.92 ± 0.11 | 79.41 ± 1.14 | 32.53 ± 0.34 | 27.81 ± 3.22 |
| Mistral-7B-v0.3 | 59.15 ± 0.10 | 35.90 ± 4.54 | 8.94 ± 1.24 | 21.60 ± 0.17 |
| Mistral-7B-v0.3-FP8 | 58.77 ± 0.13 | 32.20 ± 4.02 | 7.68 ± 1.23 | 22.72 ± 0.19 |

# Higher Temperature Amplifies Noise

At standard temperatures, original and FP8 models are statistically indistinguishable.

- Temperatures T ∈ {0.2, 0.5, 1.0, 2.0}
- Higher temperatures degrade all models' performance
- The gap between original and substitute narrows or becomes completely noisy
- Complicates statistical detection methods



# Log-Prob Fingerprinting Fails

| Model | BERT Acc | T5 Acc | GPT2 Acc | LLM2Vec Acc |
|---|---|---|---|---|
| Llama3-70B-Instruct-FP8 | 50.55 | 50.10 | 50.45 | 49.90 |
| Llama3-70B-Instruct-INT8 | 51.60 | 49.90 | 51.30 | 50.25 |
| Gemma2-9b-it-FP8 | 49.95 | 50.30 | 51.20 | 49.80 |
| Gemma2-9b-it-INT8 | 49.00 | 49.70 | 51.65 | 49.55 |
| Mistral-7b-v3-Instruct-FP8 | 50.55 | 49.75 | 48.70 | 48.75 |
| Mistral-7b-v3-Instruct-INT8 | 49.70 | 50.75 | 50.50 | 51.15 |
| Qwen2-72B-Instruct-FP8 | 50.05 | 50.25 | 48.75 | 49.80 |
| Qwen2-72B-Instruct-INT8 | 50.75 | 50.55 | 49.45 | 50.20 |

Comparing token-level log probabilities is defeated by production inference nondeterminism.

- **Software variance:** Transformers 4.40 vs 4.50, vLLM 0.6.2 vs 0.8.2
- **Hardware variance:** NVIDIA A100 vs H100
- The same exact model weights produce different log probabilities.
- Renders metadata-based detection completely unreliable.

# Randomized Substitution Attacks

Active evasion strategies make detection exponentially harder.

RANDOMIZED ROUTING

```
P_mix = (1-p)*P_spec + p*P_sub
```

Provider routes only a fraction 'p' of queries to the cheaper model. As p approaches 0, the mixed distribution converges to the specified model, requiring an impractically large query budget to detect.

BENCHMARK EVASION

```
if (is_benchmark(x)) route_to_spec()
```

Providers can detect known benchmark prompts (e.g., MMLU, GSM8K) and route them to the genuine model while serving quantized substitutes for all regular API traffic.

# Software Audits Are Fundamentally Unreliable

Black-box validation fails against realistic operational constraints.

### Statistical Tests

E.G., MMD ON TEXT OUTPUT

Requires impractically large query budgets. Fails entirely against subtle substitutions like FP8 or randomized routing attacks.

### Log-Probabilities

TOKEN-LEVEL METADATA

Defeated by natural nondeterminism in production. Changes in inference engines or GPU hardware look identical to model substitution.

### Benchmark Audits

STANDARDIZED EVALUATION

Easily bypassed by active evasion. Providers filter traffic, serving full models for tests and compressed models to users.

# The Solution: Hardware Attestation

Trusted Execution Environments (TEEs) close the verification gap.

- **Provable Integrity:** Hardware-level cryptographic attestation guarantees the exact model weights loaded and code executed.
- **Practical:** Unlike Zero-Knowledge Proofs (ZKPs) which are computationally prohibitive, TEEs offer modest overhead.
- **Robust:** Completely immune to randomized routing, benchmark evasion, and API-level obfuscation.

```
VERIFIABLE INFERENCE PIPELINE

[User] → Enclave Quote Request
[TEE] → Attestation (Code Hash, Weight Hash)
[User] → Verify Hardware Signature
[User] → Encrypted Prompt
[TEE] → Secure Execution
[TEE] → Encrypted Response
```

## Key Takeaways

- **1. Covert Substitution is Real:** Economic incentives strongly drive APIs to serve quantized/smaller models.
- **2. FP8 is a Silent Threat:** Quantization barely affects benchmarks, making it an attractive and nearly undetectable swap.
- **3. Software Fails:** Inference nondeterminism kills log-prob fingerprinting, and randomized routing defeats statistical tests.
- **4. TEEs are the Path Forward:** Hardware attestation is the only currently viable, efficient path to verifiable LLM APIs.

---

**The AI community must demand hardware-backed verifiable inference for commercial APIs.**